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3a. BFS is optimal because we're looking for the shortest path and every action costs the same amount, so the first solution found in BFS will be the shortest and therefore least expensive and best (or tied). The map start is the initial state, current coordinates are the state, goal test is if the current coordinates are the same as goal location, and the transition model is simply -1 per step with a final path cost of 40 - [# of steps].

3b. Manhattan distance works here because we're working with grid based maps and movement. Since we only have vertical or horizontal movement options, Manhattan distance is accurate.